

# HS 123: Game Theory and Economics

## BTech 5<sup>th</sup> Semester, Quiz 1

Total Marks 10, Time 55 minutes

Date: 25<sup>th</sup> Aug, 2019

[Answers should be accompanied by proper elaboration]

1. [Fiscal policy-monetary policy game] The Central Government (G) and the Reserve Bank of India (RBI) face the following hypothetical situation. The government wants to spend more money to boost the GDP. But high spending policy results in budget deficit (BD), which can lead to inflation. If G does not spend a lot, then the budget is balanced (BB). For the government BD is preferable to BB. On the other hand, the RBI can keep a high interest rate (HI) or lower the interest rate (LI). G wants the interest rate to be low, so much so that low interest in balanced budget is preferable to high interest in budget deficit. High interest policy is used by the RBI to control inflation. So, RBI prefers HI over LI in case of BD. In BB, low interest rate is preferred. For the RBI, BB with high interest rate is better than BD with high interest rate (because BB is important to it).

(a) Model the situation as a strategic game and find its Nash equilibrium/equilibria. Is the game similar to the Prisoner's Dilemma game? [3+1]

(b) Are there any dominated actions in this game? If yes, mention them. [1]

2. A mother with three children (Alia, Bikki, Chandu) discovers a broken flower pot in her house. She is more interested to find out who is the culprit, rather than punish the culprit (actually Alia broke it, the mother does not know). She asks the children to play the following game. Each child will write her/his name on a piece of paper and write "Yes, I broke it" or "No, I didn't break it" and give the paper to her. If at least one child confesses, the confessor(s) will get the normal pocket-money of Rs 20, the rest will get Rs. 50. If no one claims that she/he has broken it, then each child receives Rs. 0. Model the situation as a strategic game and write down the payoff table. Find all the Nash equilibria. Which is the focal point according to you? [2+2+1]

WJ, BD

LI > HI

1<sup>st</sup>

BR(BD) = HI

BR(BB) = LI

BR(HI) = BD

BR(LI) = BD

G

G BD > BB

G (BB, LI) > (BD, HI)

RBI HI > LI (BD)

RBI (BB, HI) > (BD, HI)

(BB, HI) > (BD, LI)  
(BB, LI) > (BB, HI)

RBI

|    | LI   | HI   |
|----|------|------|
| BB | 2, 3 | 0, 2 |
| BD | 3, 1 | 4, 3 |